NxM Templates for SuperCDMS SNOLAB Run 4 — Ge Activation Data

K-line event selection · free-pretrigger two-exponential fits · 1x1 and NxM (PCA) templates

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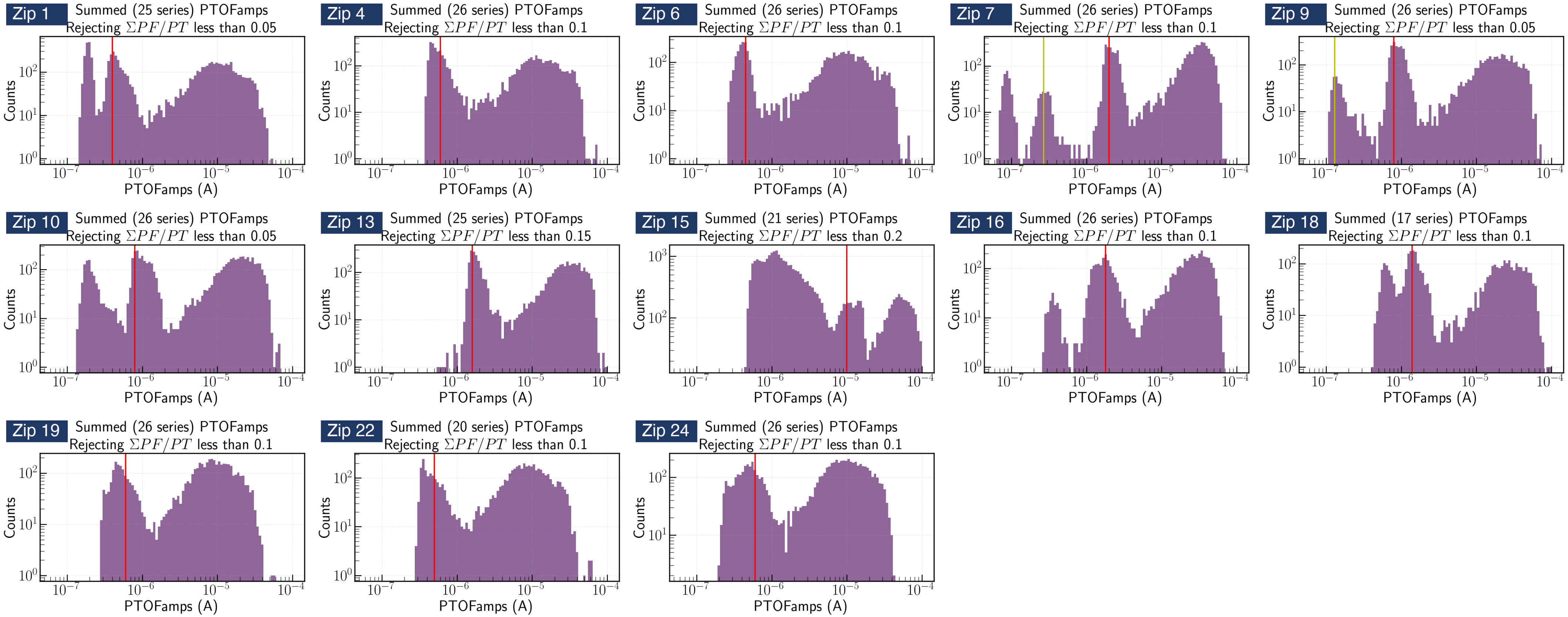
From the Ge-activation K-line to an event sample

Data: Ge Activation Data — Ops Shift 260612
[Credit: Prof. Tarek Saab · SuperCDMS Confluence]

Goal: one clean, minimally-biased pulse population per detector × channel

- Per detector, Prof. Saab's study marked the 10.37 keV K-line position in PTOFamps — the red line in each panel below
- Our event selection: a PTOFamps window bracketing the red line — position ×/÷ 1.35, a rough, somewhat arbitrary eyeballed choice; deliberately minimal, no other cuts
- For every selected event: cache the fully unprocessed raw MIDAS traces of all channels + event metadata
- 13 detectors (zips) × 27–30 series → ~120 GB raw cache

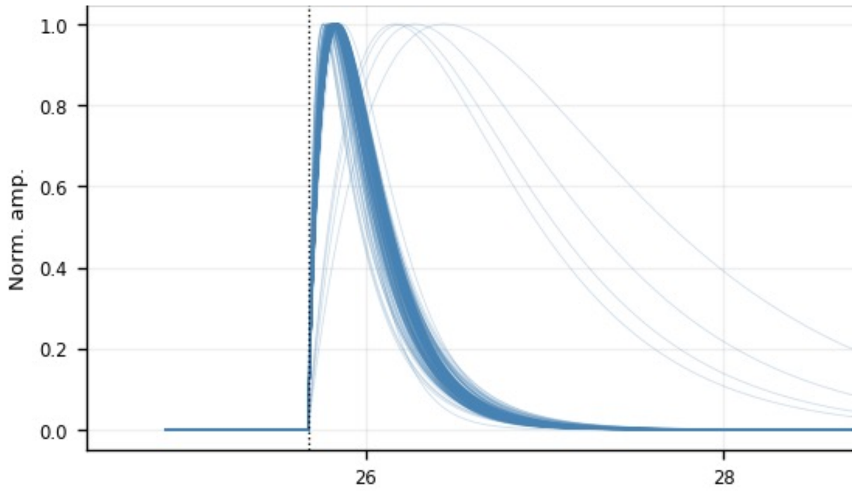


All 13 detectors, PTOFamps spectra (Prof. Saab's study)

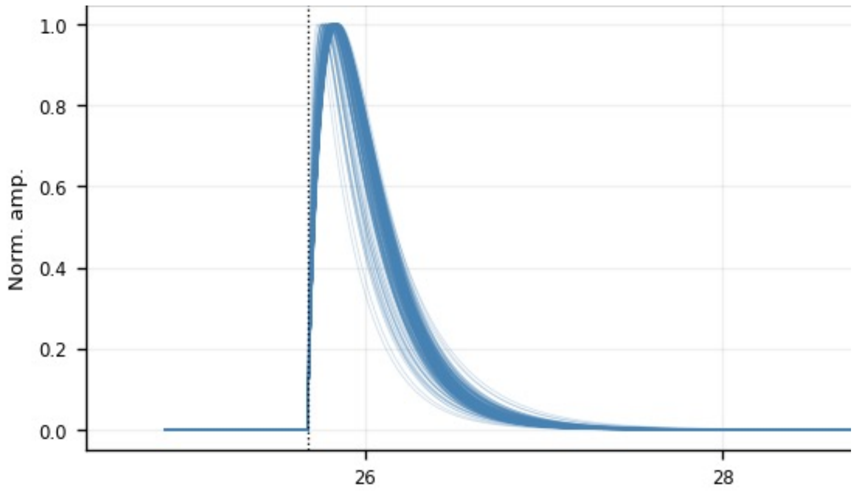
Per-trace algorithm: low-pass → fit → align

1. **Low-pass** — 100 kHz 4th-order Butterworth
2. **Normalize** — subtract the baseline, scale the trace to unit peak
3. **Fit** — two-exponential, all 5 parameters free, including the pretrigger offset:

   $y(t) = A \cdot [\exp(-(t-t_0)/\tau\_fall) - \exp(-(t-t_0)/\tau\_rise)] + b$, t_0 = pretrigger ∈ 16050 ± 3000 samples
4. **Quality numbers** — fit_ok := A>0 and 0<τ_rise<τ_fall; NRMSE := RMS(residual)/fit peak
5. **Align** — shift the measured trace by (fitted pretrigger − 16050)



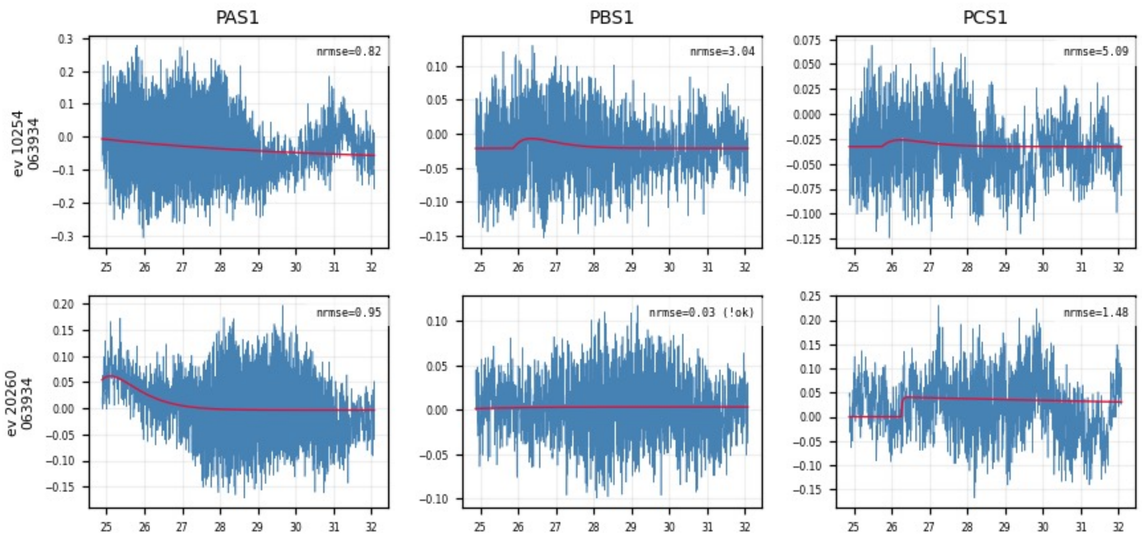
All fit_ok fitted curves at the common pretrigger, peak-normalized — zoom on the pulse (Z7 PBS1, time in ms)



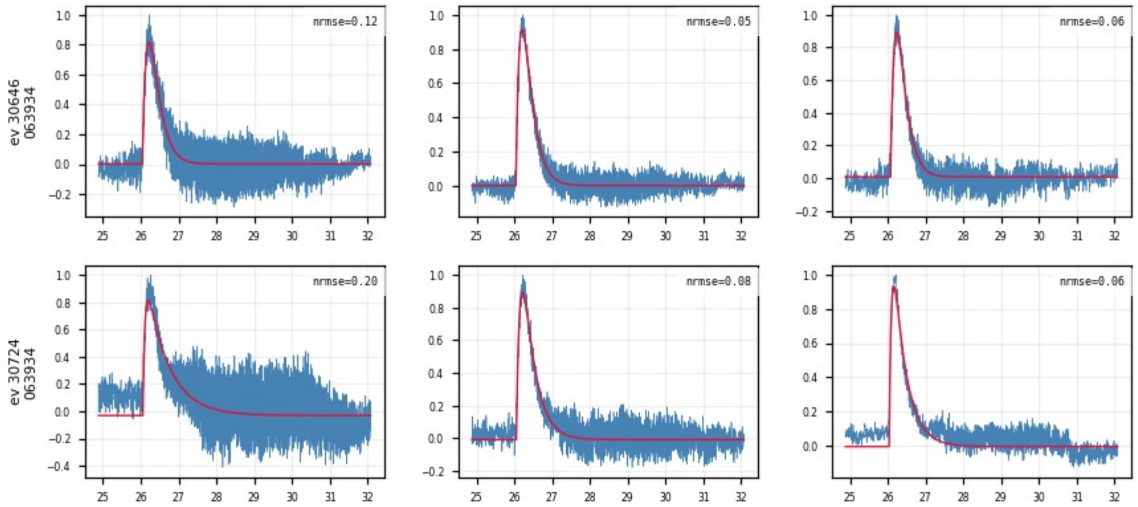
Same, after NRMSE ≤ 0.4 — a single tight shape family remains

Fit quality at a glance — event × channel grids

- One row per event, one column per channel: low-passed trace (blue) vs 2-exp fit (red), NRMSE stamped per panel
- The same event fits consistently across channels; a noise trigger fails in all channels at once
- Every figure carries its full processing chain in the header → self-documenting



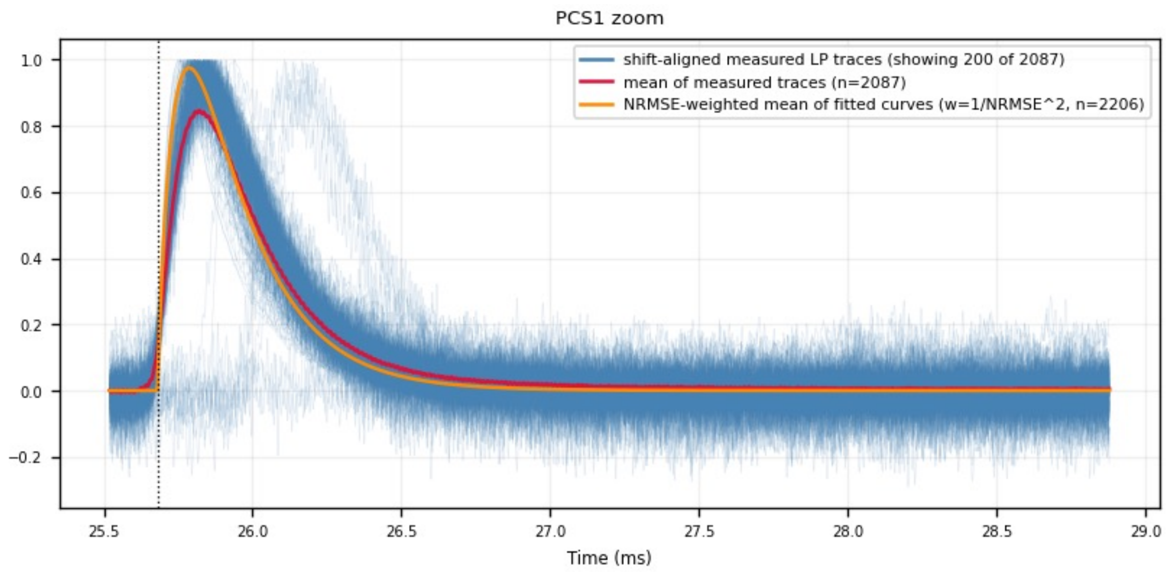
Noise triggers: the fit fails in every channel at once



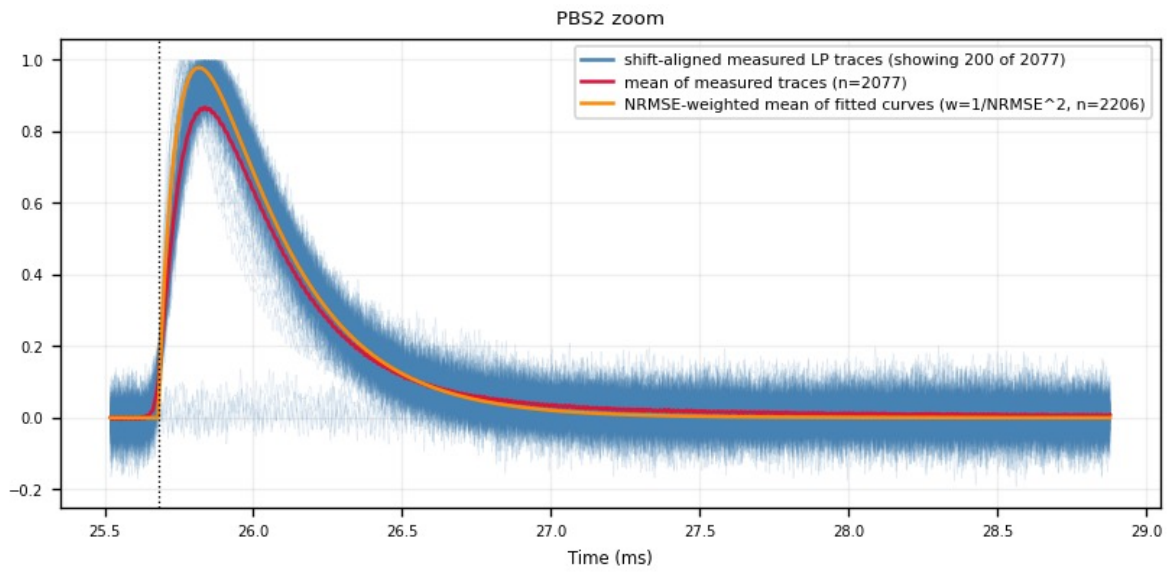
K-line events: consistent good fits across channels

Alignment result — overlaid measured traces

- All fit_ok traces shifted to the common pretrigger (blue) + point-by-point mean (red) + NRMSE-weighted mean of the fitted curves (orange, w = 1/max(NRMSE, 0.01)²)
- Peak zoom on a quiet detector: a tight bundle — the template input is well defined. Note the faint displaced bundle beside the main one, visible on both crystal faces (S1 and S2) — more on it later



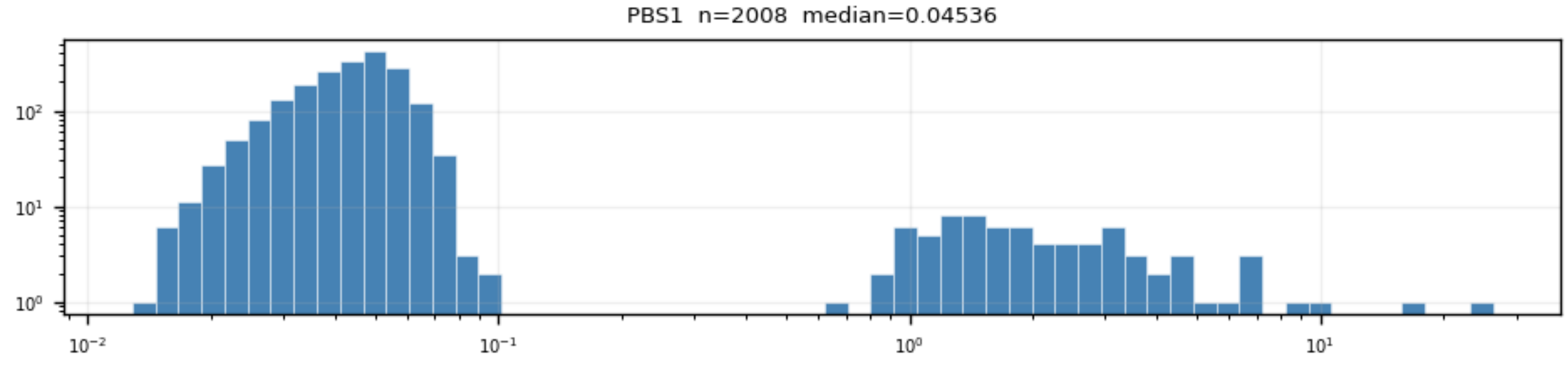
Z7 PCS1 (S1 face) — peak zoom; the faint displaced bundle is clearly visible



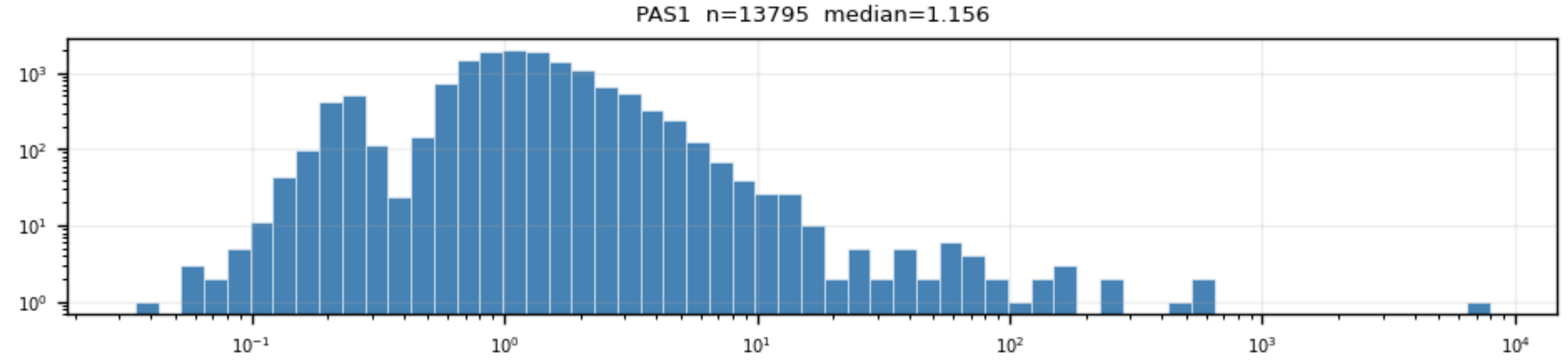
Z7 PBS2 (S2 face) — same faint displaced bundle

Quality cut 1: NRMSE ≤ 0.4 — where the number comes from

- NRMSE of fit_ok events is bimodal: good fits (median ≈ 0.05–0.1) vs noise triggers (≈ 1–2), valley at ≈ 0.4–0.5
- The cut sits in the valley — it is read off the distribution, not tuned on the templates
- Weak detectors (Z1, Z4, Z6, Z18, Z19, Z22, Z24): the PTOF window admits a noise-dominated mixture — this cut is what extracts the real-pulse population



Z7 PBS1 (quiet): two clean populations, log-log axes

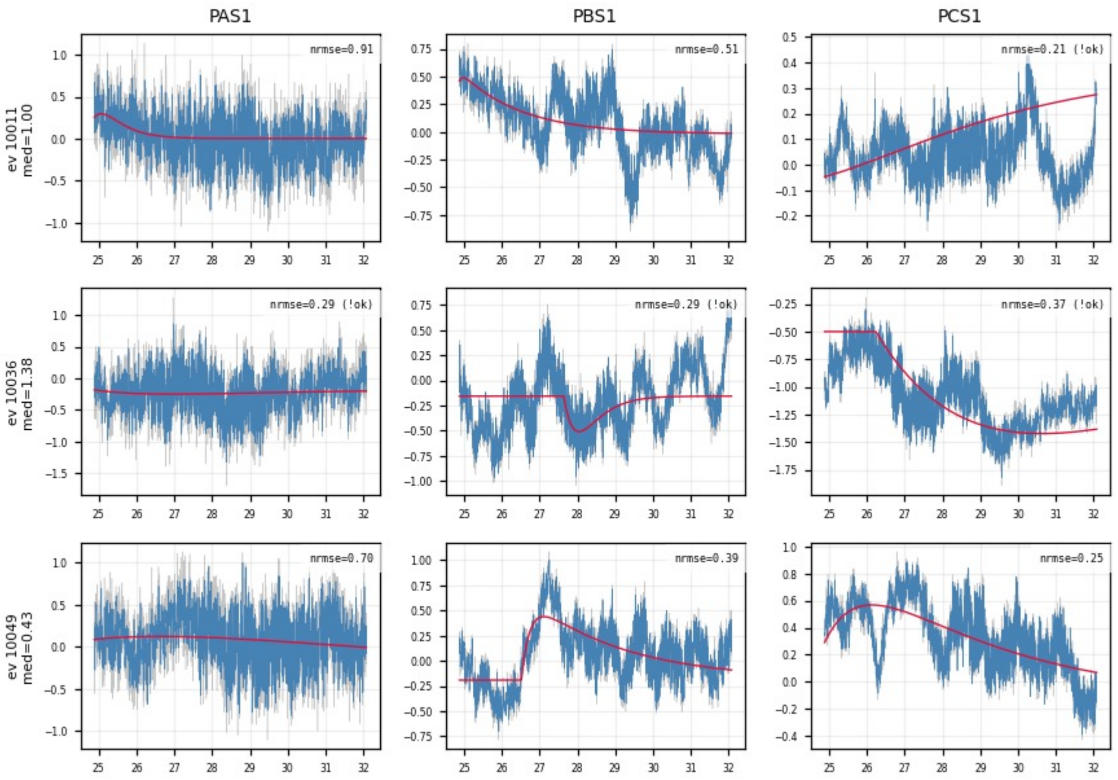


Z22 PAS1 (weak): noise population dominates — the window alone is not enough

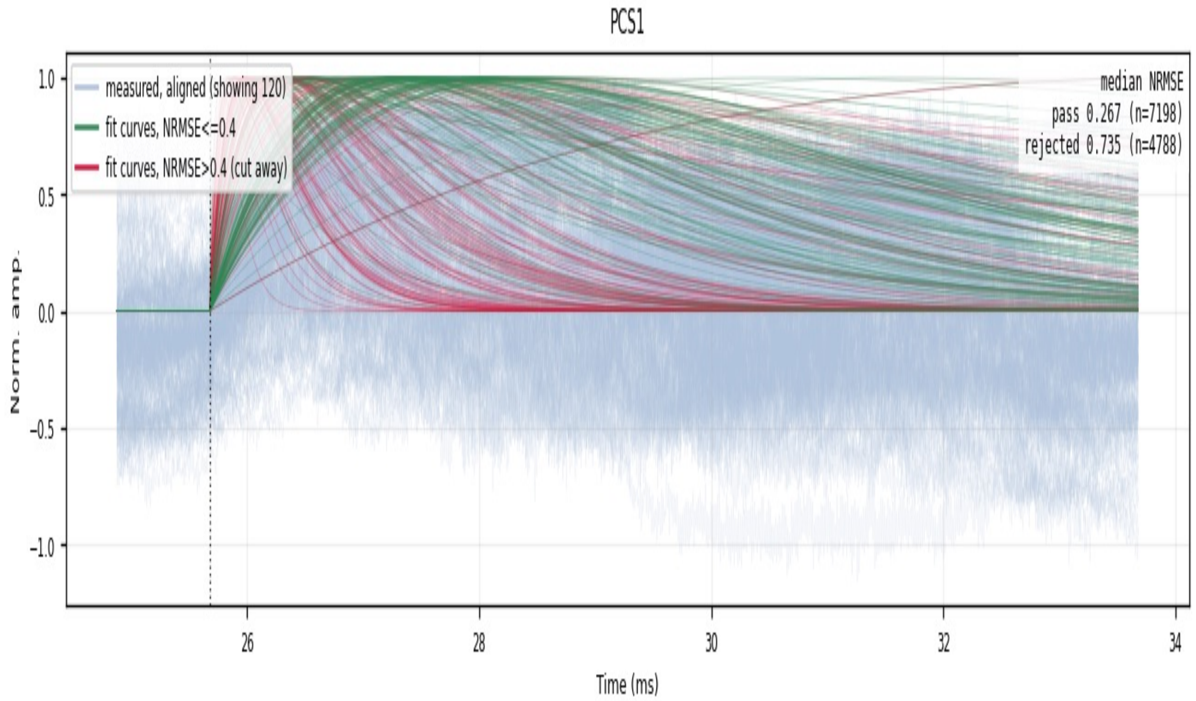
The rejected population is noise — verified in the raw traces

• Event grids of NRMSE-rejected events (median NRMSE > 0.4 across channels): the raw traces show no pulse — the cut removes noise triggers, not physics

• Fan-cut view: fitted curves that pass (green) vs cut away (red) on top of the aligned data, median NRMSE of each population stamped per channel



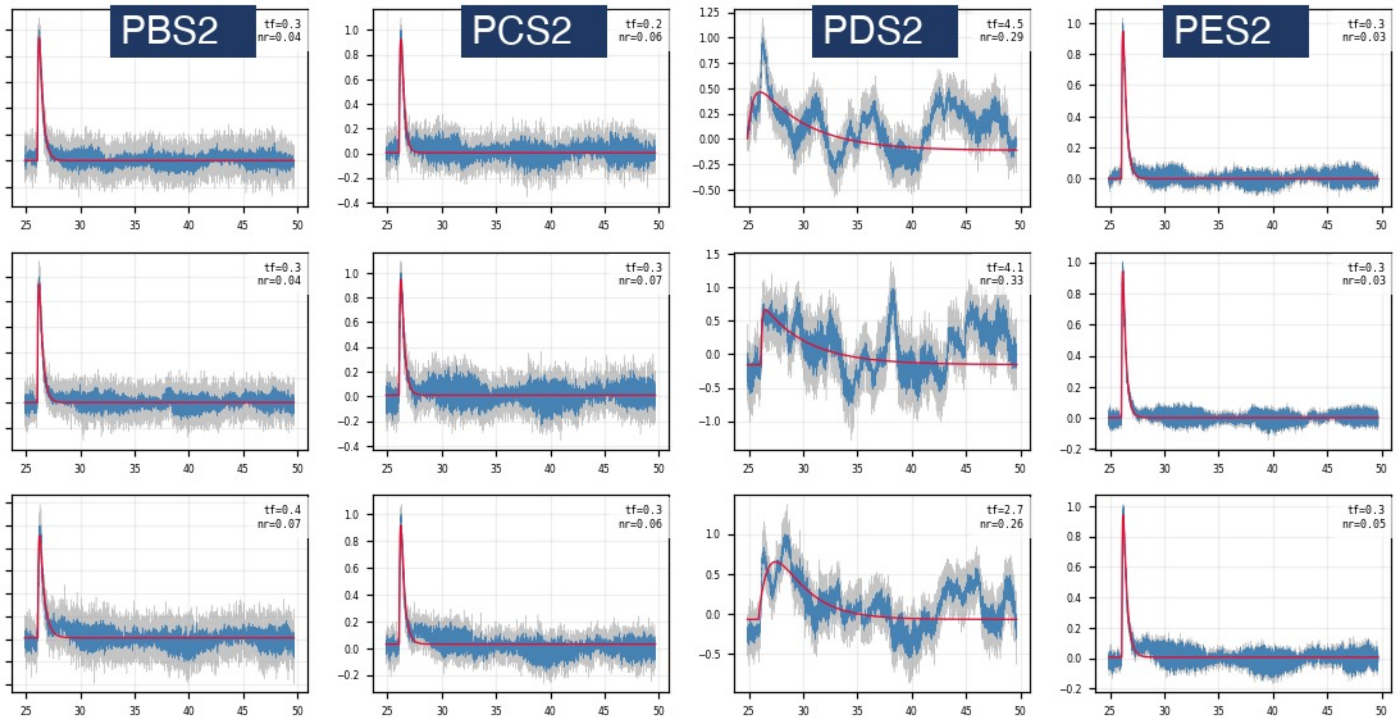
Z22: NRMSE-rejected events — raw traces are noise



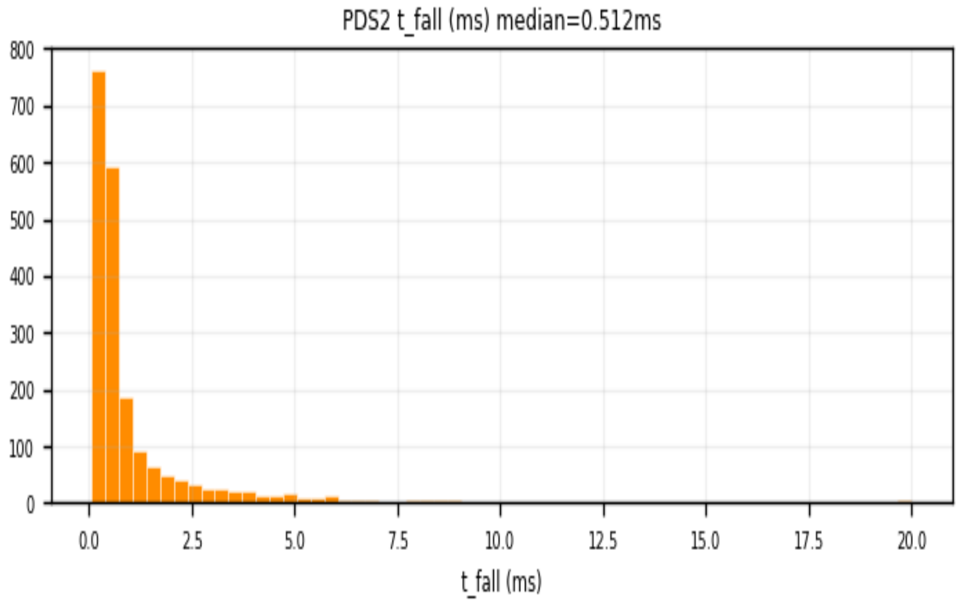
Z22 PCS1: pass (green) vs rejected (red) — vertically stretched for visibility

Follow-up 1 — the slow-fall tail is a one-channel artifact

- The post-cut fan still shows slow-fall tails → sample them: NRMSE ≤ 0.4 AND τ_fall > 1.5 ms, 10 random events, raw vs fit drawn in all 12 channels
- The sampled events are real pulses → kept. Their extreme fall times trace to one channel: only PDS2 swings (τ_fall median 0.51 ms vs ≈ 0.25 ms elsewhere) — a low-frequency disturbance
- Conclusion: no τ_fall cut — slow-fall events passing NRMSE stay in; the extreme tail is a one-channel artifact, not physics



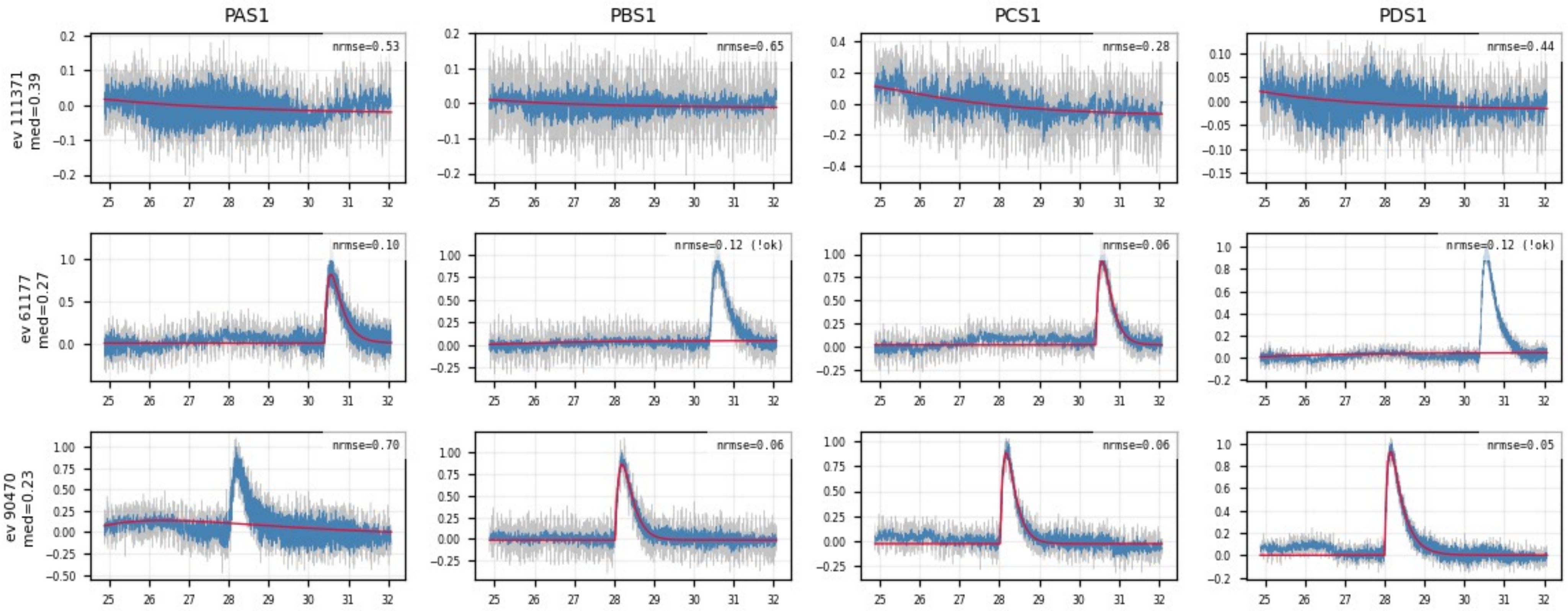
Z7, 3 of the sampled slow-fall events: normal pulses in PBS2/PCS2/PES2, low-frequency swings only in PDS2



Z7 PDS2, fitted τ_fall: broad tail (median 0.51 ms) — the artifact in the distributions

Follow-up 2 — genuine slow-rise pulses (“shadow” events)

- Same sampling recipe on the rise side: median NRMSE ≤ 0.4 AND median τ_rise > 0.2 ms → the raw traces show real pulses — pretrigger aligned, peak late, consistent across channels
- They are the faint displaced bundle in the aligned overlays — a genuine second pulse shape (candidate surface/bulk effect, not settled)
- Real physics → cannot simply be cut away; exactly the shape variation the multi-template NxM method is built to capture



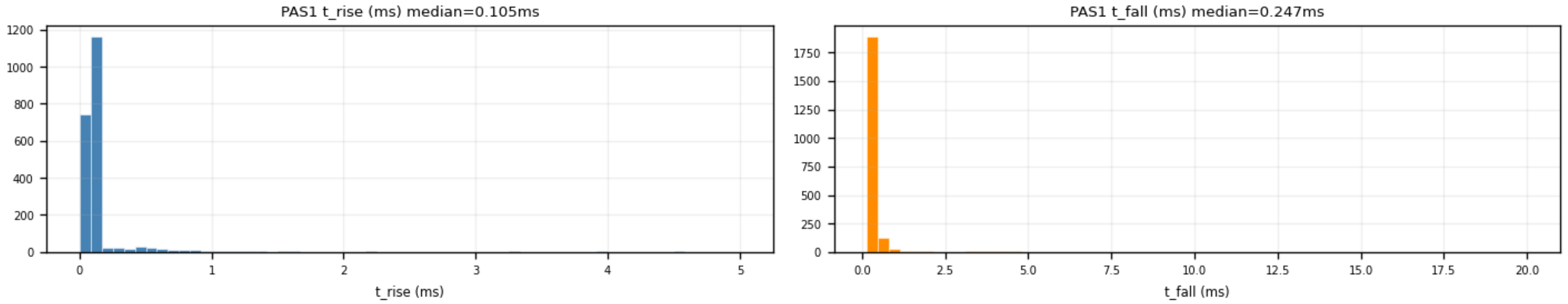
Z7 shadow events (first 4 channels): raw (gray) / LP (blue) / fit (red) — genuine slow pulses, pretrigger aligned, peak late

Slow drift also fits “slow”: a τ_rise ≤ 0.3 ms ceiling

• On noisy-window detectors a smooth baseline drift survives the NRMSE cut — a slow 2-exp hugs it with a tiny residual — and mimics a slow rise

• Fast pulses sit at τ_rise ≈ 0.1 ms (p90 ≈ 0.15 ms), far below the drift tail → a ceiling at 0.3 ms blocks the drift at ≈ 2–5% signal cost

• Keeps most genuine slow pulses, trims the very slowest — documented trade-off, final decision pending

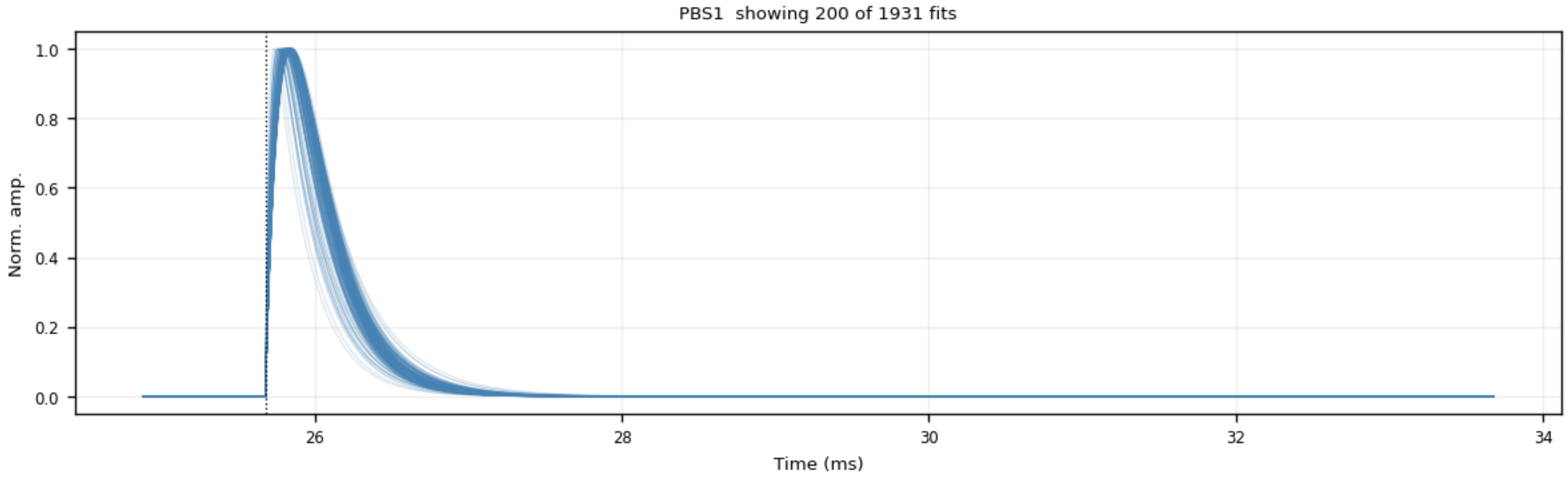


Z7 PAS1: fitted τ_rise (median 0.105 ms) and τ_fall distributions

• On quiet detectors both τ_rise and τ_fall distributions are narrow — the pulse shape is stable across events

Template family 1 — analytic 2-exp, NRMSE-weighted (1x1)

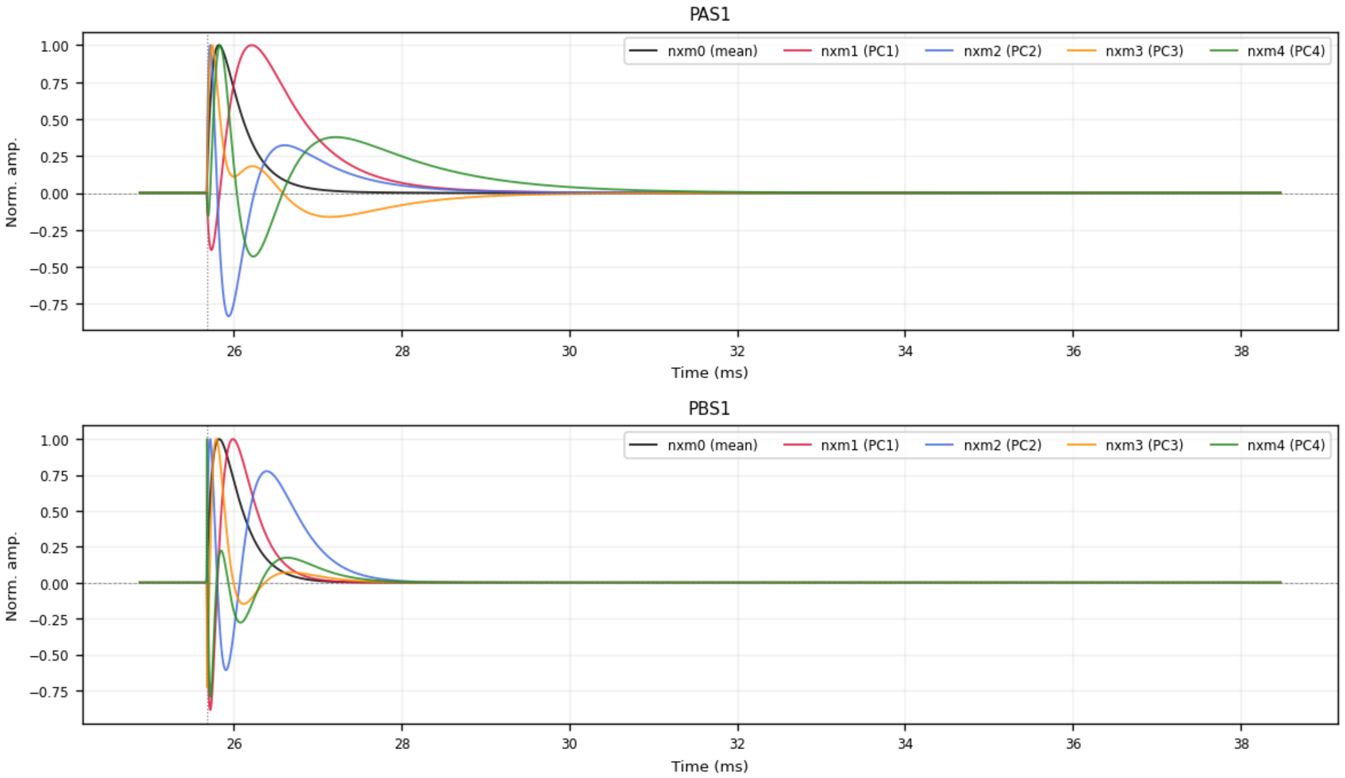
- Per channel: weighted mean of ALL fit_ok fitted curves at the common pretrigger, weight $w = 1/\max(\mathrm{NRMSE}, 0.01)^2$
- Badly-fit events count less but are never excluded — robust and smooth (noise-free by construction)
- Delivered as peak-normalized 32768-bin ROOT TH1D per channel + summed PT/PS1/PS2 templates



Z7 PBS1, fitted curves after NRMSE ≤ 0.4 and τ_rise ≤ 0.3 ms — the NRMSE-weighted mean of this family is the 1x1 template

Template family 2 — NxM PCA templates

- Input: fitted curves passing fit_ok + NRMSE ≤ 0.4 + τ_rise ≤ 0.3 ms, at common pretrigger, peak-normalized (PCA window 15550–24050, ≤ 3000 curves/channel)
- nxm0 = mean shape; nxm1…nxm4 = first four principal components — a real pulse is fit as $\Sigma_i \, amp_i \cdot nxm_i$
- PC1 + PC2 already capture 96–98% of the shape variance; final step before delivery: normalize all five to unit peak — the delivered product



Z7 PAS1 / PBS1: nxm0 (mean) + nxm1–4 (PCs) — the oscillating components encode rise/fall-time variation

Delivered — and what remains

- Delivered for all 13 zips, official cdmsbats PulseTemplates layout (zip{N}/{chan}, {chan}nxm0–4, summed PT/PS1/PS2):
  - SNOLAB_R4_20260706_ZhihengLi_zip{N}.root — 2-exp NRMSE-weighted (1x1)
  - SNOLAB_R4_20260707_ZhihengLi_pca_zip{N}.root — NxM PCA (normalized)
- Every step is traceable: raw cache → per-series fit checkpoints indexed by EventNumber → self-documenting figures (11 types × 13 zips, all versioned)
- Cuts derived from the data, validated in raw traces: NRMSE ≤ 0.4 (bimodal valley), τ_{rise} ≤ 0.3 ms (drift leakage)
- Open items:
  - final decision on the τ_{rise} ceiling vs the genuine slow-pulse population
  - run the group's template-validation method on the new templates